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A granular XGBoost classification algorithm

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Abstract

This paper proposes the first integration of multi-distance granular computing with XGBoost, significantly improving generalization in small-sample scenarios through enriched feature representations. XGBoost (eXtreme Gradient Boosting) is a machine learning algorithm primarily utilized to address classification and regression problems. It typically relies on the original features of the data or basic feature engineering, which does not fully exploit the internal structures and correlations inherent in the data. Meanwhile, due to the problems of data sparsity, category imbalance, and the presence of noise in small sample datasets, its performance often deteriorates, leading to a decrease in the model's generalization ability. In this paper, we propose a granular XGBoost classification algorithm designed to enhance its classification accuracy in small-sample datasets. The algorithm initially extends the dimensionality of the dataset by combining multiple features and subsequently employs various distance metrics for granulation processing, generating multiple granularity levels of data representation, resulting in richer and more diverse feature representations. The feature representations at different granularity levels are then fused separately and fed into the XGBoost classifier for training and prediction. This approach to mitigate the bias and error associated with a single metric, thereby achieving a better balance in the representations of each category within the dataset. Experimental results indicate that the classification accuracy of the granular XGBoost classification algorithm, which utilizes the multi-distance metric method, is significantly improved across various datasets when compared to the traditional XGBoost algorithm but also reinforces its robustness and generalization capability across different datasets and feature distributions, providing novel insights and methodologies to address the overfitting challenges associated with small sample data.

Keywords XGBoost · Granular computing · Multiple distances · Classification

1 Introduction

In 1979, Zadeh [1] introduced the concept of information granularity, which ignited significant interest in academia and established the groundwork for the development of granular computing. In 1982, Pawlak [2] proposed the rough set model, which fundamentally aims to characterize a set with fuzzy boundaries by partitioning the complete set into two precise subsets: the upper approximation set and the lower approximation set. In 1985, Hobbs [3] examined the decomposition and merging of granules, presenting methods and models for generating granules of varying sizes. In 1996, during his visit to UC Berkeley, Lin [4] formally articulated the concept of granular computing. The following year, Zadeh [5] further underscored the significance of processing information through granules at multiple levels, emphasizing the multi-level and multi-granularity aspects of information processing. Lin also investigated the model of granular computing within binary relations, offering a comprehensive analysis of granular structures, granular descriptions, and the practical applications of granules [6–8]. Building on Lin's work, Yao [9–11] conducted comprehensive research on granular computing by integrating the neighborhood system and applying it to knowledge discovery. He established the relationship between if-then rules and granular sets, and proposed a method for solving consistent classification problems using the lattice formed by all partitions, thereby

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offering new methods and perspectives for data mining. Bu et al. [12] analyzed clustering and classification techniques from the perspective of information granularity, striving to unify these processes under the framework of information granularity principles. They highlighted that, from the viewpoint of information granularity, clustering is performed at a uniform granularity, whereas classification occurs at varying granularities. These studies have revitalized the field of granular computing, enriched its theoretical framework, and introduced new perspectives and methods for addressing practical problems, thus advancing research and development in data science [13], machine learning [14–16], and artificial intelligence [17, 18].

XGBoost is a machine learning algorithm based on gradient boosting decision trees, proposed and open-sourced by Chen and Guestrin [19]. Through continuous optimization and extension, XGBoost has achieved significant performance improvements by refining the tree-building algorithm, incorporating regularization terms, and optimizing parallel computation, making it effective for large-scale datasets. XGBoost has not only frequently won numerous data science competitions but has also been widely adopted in both industry and academia, becoming the preferred algorithm for various tasks such as classification, regression, and ranking. Due to its performance advantages, XGBoost is extensively utilized in diverse fields, including risk prediction [20], risk assessment [21–23], and fault diagnosis [24–26]. However, its limitations become pronounced in small-sample scenarios, where raw or simplistic feature engineering often inadequately captures the underlying data structures and multi-scale correlations, and the algorithm's inherent sensitivity to noise and category imbalance frequently induces overfitting. Liu et al. [27] proposed a hybrid inversion model that integrates XGBoost with deep convolutional neural networks (DCNNs), which optimizes rice biomass estimation under limited sample conditions through simulated data generation. Li et al. [28] proposed a knowledge-enhanced XGBoost method that combines wavelet entropy features, CNN learning, attention fusion, and semi-supervised self-training, achieving superior fault diagnosis performance under small-sample conditions. Yeşilkanat and Akkoyun [29] proposed a framework combining SMOTE-based data augmentation with an XGBoost classifier to address class imbalance in neutron-halo classification tasks.

According to granular computing theory [13], multi-granularity reduces the model variance and avoids over-reliance of the model on the local noise of a small number of samples through the fusion of features with different granularities. Therefore, this paper proposes a granular XGBoost classification algorithm (GXGBoost), which employs various distance measures (such as Euclidean, Manhattan, and Chebyshev distances) to granularize the dataset after executing multi-feature combinations. This methodology produces a feature representation that is both multi-level and multi-perspective. The primary objective of the multi-feature combination technique is to elevate the original low-dimensional features into higher dimensions by utilizing a pairwise combination method on the outcomes. This procedure aids in extracting additional feature information from the existing data, effectively tackling the challenges of low dimensionality while reducing the likelihood of the model overfitting the training dataset. At the same time, the granularization process amalgamates the granular representations obtained from different distance metrics, creating a dataset that is rich in multi-level features. This, in turn, offers a more thorough data representation and boosts the model's learning capacity and generalization ability. In addition, the proposed approach indirectly enhances the robustness of XGBoost on imbalanced datasets. By generating granular representations based on multiple distance metrics, the minority class data points gain more discriminative representation in the transformed feature space. This leads to better recognition performance without requiring explicit imbalance handling strategies such as resampling or cost-sensitive learning.

The remainder of this paper is organized as follows: Section 2 introduces granular computing fundamentals. Section 3 details the multi-distance granularization methodology. Section 4 presents the GXGBoost algorithm. Section 5 validates the approach through extensive experiments. Section 6 concludes the work.

2 Granules and granular computing

2.1 Granules and granular vectors

Granularization refers to the process of decomposing complex data or information into smaller, more manageable units, referred to as granules, at varying levels of detail. This approach enhances understanding and facilitates efficient handling of the data. Granules may consist of fundamental units or more abstract concepts, depending on the context. When necessary, a structured, constrained, and repeatable collection is designated as a granule, which is accompanied by definitions for granular vectors and a set of rules governing the management of these granules.

Definition 1 Suppose there is a set $g = \{r_1, r_2, \ldots, r_n\}$, where each element $r_i \in R$ and each r_i is referred to as a granular core, the g is called a granule [14]. This granule can be represented as:

$$g = \{r_j\}_{j=1}^{n}. \tag{1}$$

Definition 2 If each granule g_i is composed of a set $\{r_j\}_{j=1}^{n}$, a vector consisting of multiple such granules is defined as a granular vector [14]. An m-dimensional granular vector can be represented as $G = \{g_1, g_2, \ldots, g_m\}^T$, and it can also be expressed as:

$$G = \{g_i\}_{i=1}^{m}. \tag{2}$$

2.2 Granule operations

Granular computing theory emphasizes the importance of granules, which serve as a fundamental component of its framework. These granules facilitate data processing at various levels and degrees of granularity. The operations involved are not limited to conventional set theory approaches; rather, they can be adapted based on the nature of the granules or specific requirements. Fundamental operations such as addition, subtraction, multiplication, and division are also examined and applied within the context of granular computing. This formalization of the relationships and interactions among granules allows for the development of mathematical frameworks capable of processing information across different granularities and at varying levels of complexity.

Definition 3 Suppose there are two granules, g and h, where each granule is a set of numerical values, *i.e.* $g = \{r_1, r_2, \ldots, r_n\}$ and $h = \{h_1, h_2, \ldots, h_n\}$, and all elements in granule h are non-zero. Then, the addition, subtraction, multiplication, and division operations [14] between the two granules are defined as:

$$g + h = \{r_j + h_j\}_{j=1}^{n}, \tag{3}$$

$$g - h = \{r_j - h_j\}_{j=1}^{n}, \tag{4}$$

$$g * h = \{r_j * h_j\}_{j=1}^{n}, \tag{5}$$

$$g / h = \{r_j / h_j\}_{j=1}^{n}. \tag{6}$$

Definition 4 Suppose there is a granule g which is a numerical set, i.e. $g = \{r_1, r_2, \ldots, r_n\}$. Define a mapping $f{:}g \to g'$, which associates each element r_i in the granule g with a real-valued result. Then the functional operation of a granule [14] is defined as:

$$f(g) = \{f(r_j)\}_{j=1}^{n}. \tag{7}$$

Example 1 The softmax and Relu operations for a granule g are defined as follows:

$$\operatorname{softmax}(g) = \left\{ \frac{e^{r_i}}{\sum_{j=1}^{n} e^{r_j}} \right\}_{i=1}^{n}, \tag{8}$$

$$\operatorname{Relu}(g) = \{\max(0, r_j)\}_{j=1}^{n}. \tag{9}$$

2.3 Granular vector operations

In the context of granular computing, operations such as addition, subtraction, multiplication, division, and dot product concerning granular vectors are fundamental to data processing. Unlike conventional vector algebra which emphasizes numerical computation efficiency, granular vector operations in our method aim to enrich the semantic representation of features by aggregating multi-distance-based perspectives. Therefore, the granularity vector construction focuses on enhancing feature discrimination rather than computational optimization. These operations enhance the applicability of granular vectors in mathematical analyses. Simultaneously, they establish a foundational framework for data merging, comparative evaluations, and more complex interactions, although the specific applications may vary widely across different scenarios.

Definition 5 Suppose that given two m-dimensional granular vectors as $G = \{g_1, g_2, \ldots, g_m\}^T$ and $H = \{h_1, h_2, \ldots, h_m\}^T$, where all elements of the granules in the granular vectors H elements are non-zero. Then, the addition, subtraction, multiplication, and division operations [14] between the two granular vectors are defined as:

$$G + H = \big(g_1 + h_1, \ldots, g_m + h_m\big), \tag{10}$$

$$G - H = \big(g_1 - h_1, \ldots, g_m - h_m\big), \tag{11}$$

$$G \times H = \big(g_1 \times h_1, \ldots, g_m \times h_m\big), \tag{12}$$

$$G / H = \big(g_1 / h_1, \ldots, g_m / h_m\big). \tag{13}$$

Definition 6 Suppose that given two m-dimensional granular vectors as $G = \{g_1, g_2, \ldots, g_m\}^T$ and $H = \{h_1, h_2, \ldots, h_m\}^T$, then the dot product operation [14] between the two granular vectors is defined as:

$$G \bullet H = g_1 * h_1 + g_2 * h_2 + \cdots + g_m * h_m. \tag{14}$$

The dot product between two granular vectors produces a granule, which quantifies the degree of similarity between the two vectors within the vector space. A larger granule value typically indicates that both vectors exhibit high values across multiple dimensions, suggesting a stronger directional similarity. Conversely, a granule value of zero generally indicates that the two vectors are independent within the vector space and exhibit no similarity.

3 Multi-distance metric granularization

Multi-distance metric granularization is a complex technique primarily used in the fields of data analysis and machine learning. This method involves breaking down and partitioning data into segments, utilizing various distance measures to gain a deeper understanding of the intricate interrelations among the data. In this approach, data is divided into granules, which represent distinct subsets or segments of the data space features. Each granule is subsequently analyzed using different types of distance measures, such as Euclidean, Manhattan, and Chebyshev, among others. The analysis then focuses on comparing these granules to assess their similarities and differences.

Give a multi-distance granularization system $M = (X, F, P, L, D)$, where the sample set is $X = \{x_1, x_2, \ldots, x_n\}$, the feature set is $F = \{f_1, f_2, \ldots, f_m\}$, the reference sample set is $P = \{p_1, p_2, \ldots, p_q\}$, the decision set is $L = \{l_1, l_2, \ldots, l_n\}$, and the set of distance methods is $D = \{d_1, d_2, \ldots, d_k\}$. Each distance threshold $d_i \in D$ is used to define different granular scales. For a given feature set $F = \{f_1, f_2, \ldots, f_m\}$, and two features f_a, f_b are paired to form a feature plane coordinate system $< f_a, f_b >$or$< f_b, f_a >$. The first element is considered the *x*-coordinate in the 2D space, and the second element is considered the *y*-coordinate.

3.1 Granular distance metrics

Granular distance metrics play a crucial role in both granular computing and data analysis. They are employed to understand, analyze, and explore the internal structure and relationships within data. By measuring the distance or similarity between granules, this method provides valuable insights. In granular computing, granules are regarded as fundamental data units, and distance metrics offer a means to quantify the relationships that exist between these units. This quantification facilitates various tasks, such as clustering, classifying, and conducting analyses based on data similarity.

Definition 7 Given a single sample $x \in X$, a reference sample $p \in P$, a combination of features $(c_1, c_2) \in C$ and a single distance granularization method $d \in D$, then *x* with *p* on the feature (c_1, c_2) using the granulation method *d* the result obtained after is:

$$d(x, p) = v(x_{<f_{c_1}, f_{c_2}>}, p_{<f_{c_1}, f_{c_2}>}), \tag{15}$$

where *v* is the distance between *x* and *p* on feature (c_1, c_2) calculated using granularization method *d*.

Definition 8 Given a granule $g = \{r_1, r_2, \ldots, r_n\}$, two granule core r_i, r_j are arbitrarily chosen for two-by-two combinations, and combined to form a combined granule $g_1 = \{\langle r_1, r_2\rangle, \langle r_1, r_3\rangle, \cdots, \langle r_{n-1}, r_n\rangle\}$.

Definition 9 Given two combined granules $g = \{\langle r_{a1}, r_{b1}\rangle, \langle r_{a2}, r_{b2}\rangle, \cdots, \langle r_{an}, r_{bn}\rangle\}$ and $h = \{\langle h_{a1}, h_{b1}\rangle, \langle h_{a2}, h_{b2}\rangle, \cdots, \langle h_{an}, h_{bn}\rangle\}$, the multiple distance metric methods for several granules are defined as:

Manhattan Distance:

$$D_h(g, h) = \sum_{j=1}^{n} |r_{aj} - h_{aj}| + \sum_{j=1}^{n} |r_{bj} - h_{bj}|, \tag{16}$$

Euclidean distance:

$$D_e(g, h) = \sum_{j=1}^{n} \sqrt{(r_{aj} - h_{aj})^2 + (r_{bj} - h_{bj})^2}, \tag{17}$$

Squared Euclidean distance:

$$D_q(g, h) = \sum_{j=1}^{n} (r_{aj} - h_{aj})^2 + \sum_{j=1}^{n} (r_{bj} - h_{bj})^2, \tag{18}$$

Cosine distance:

$$D_c(g,h)=1-\frac{\sum_{j=1}^{n}\left(r_{aj}*h_{aj}\right)+\sum_{j=1}^{n}\left(r_{bj}*h_{bj}\right)}{\sqrt{\sum_{j=1}^{n}\left(r_{aj}*r_{aj}+h_{aj}*h_{aj}\right)}}*\frac{1}{\sqrt{\sum_{j=1}^{n}\left(r_{bj}*r_{bj}+h_{bj}*h_{bj}\right)}}, \tag{19}$$

Chebyshev Distance:

$$D_{cb}(g,h)=\max(\sum_{j=1}^{n}\left|r_{aj}-h_{aj}\right|,\sum_{j=1}^{n}\left|r_{bj}-h_{bj}\right|), \tag{20}$$

Canberra Distance

$$D_{ca}(g,h)=\sum_{j=1}^{n}\frac{|r_{aj}-h_{aj}|}{|r_{aj}|+|h_{aj}|}+\sum_{j=1}^{n}\frac{|r_{bj}-h_{bj}|}{|r_{bj}|+|h_{bj}|}, \tag{21}$$

Bray Curtis Distance:

$$D_b(g,h)=\frac{\sum_{j=1}^{n}|r_{aj}-h_{aj}|+\sum_{j=1}^{n}|r_{bj*}h_{bj}|}{\sum_{j=1}^{n}\left(|r_{aj}|+|r_{bj}|+|h_{aj}|+|h_{bj}|\right)}, \tag{22}$$

Ochiai distance:

$$D_o(g,h)=\frac{\sum_{j=1}^{n}\left(r_{aj}*h_{aj}\right)+\sum_{j=1}^{n}\left(r_{bj}*h_{bj}\right)}{\sqrt{\sum_{j=1}^{n}\left(r_{aj}*r_{aj}+r_{bj}*r_{bj}\right)}}*\frac{1}{\sqrt{\sum_{j=1}^{n}\left(h_{aj}*h_{aj}+h_{bj}*h_{bj}\right)}}, \tag{23}$$

Product distance:

$$D_p(g,h)=\sum_{j=1}^{n}\left(r_{aj}*h_{aj}\right)+\sum_{j=1}^{n}\left(r_{bj}*h_{bj}\right), \tag{24}$$

Table 1 Dataset

X	f_1	f_2	f_3	L
x_1	0.3	0.5	0.6	0
x_2	0.5	0.2	0.4	1
x_3	0.7	0.8	0.5	0

where r_{aj}, r_{bj}, h_{aj} and h_{bj} denote the values of combined granules g and h in the $j-th$ dimension, respectively.

Example 2 Given a dataset $U=(X,F,L)$, where the sample set is $X=\{x_1,x_2,x_3\}$, the feature set is $F=\{f_1,f_2,f_3\}$ and the decision set is $L=\{l_1,l_2,l_3\}$, as shown in Table 1. For this dataset, sample x_1 is selected as the reference sample $P=(0.5,0.2,0.4)$, and multiple distance metrics, such as the Manhattan distance and the Euclidean distance, are applied for granularization.

For sample x_2, given the feature combination reference sample $P=\{<0.5,0.2>,<0.5,0.4>,<0.2,0.4>\}$, the results obtained using the aforementioned multi-distance metric granularization methods are:

$$\begin{aligned}G_{x_1<f_1,f_2>}=&(|0.3-0.5|+|0.5-0.2|,\\&\sqrt{(0.3-0.5)^2+(0.5-0.2)^2},(0.3-0.5)^2+\\&(0.5-0.2)^2,1-\frac{0.3\cdot0.5+0.5\cdot0.2}{\sqrt{0.3^2+0.5^2+0.5^2+0.2^2}},\\&\max(|0.3-0.5|,|0.5-0.2|),\frac{|0.3-0.5|}{|0.3|+|0.5|}+\\&\frac{|0.5-0.2|}{|0.5|+|0.2|},\frac{|0.3-0.5|+|0.5-0.2|}{|0.3|+|0.5|+|0.5|+|0.2|},\\&\frac{0.3\cdot0.5+0.5\cdot0.2}{\sqrt{0.3^2+0.5^2}\cdot\sqrt{0.5^2+0.2^2}},0.3\cdot0.5+\\&0.5\cdot0.2)\\&=(0.5,0.36056,0.13,0.20384,0.3,0.67857,\\&0.33333,0.79616,0.25)\\G_{x_1<f_1,f_3>}=&(0.39999,0.28284,0.07999,0.09204,0.2,\\&0.44999,0.22222,0.90796,0.39)\\G_{x_1<f_2,f_3>}=&(0.49999,0.36056,0.12999,0.02658,0.3,\\&0.62857,0.29412,0.97342,0.33999)\end{aligned}$$

For sample x_2, given the feature combination reference sample $P=\{<0.5,0.2>,<0.5,0.4>,<0.2,0.4>\}$, the results obtained using the aforementioned multi-distance metric granularization methods are:

$$\begin{aligned}G_{x_2<f_1,f_2>}&=(0.0,0.0,0.0,0.0,0.0,0.0,0.0,1.0,0.29)\\G_{x_2<f_1,f_3>}&=(0.0,0.0,0.0,0.0,0.0,0.0,0.0,1.0,0.41)\\G_{x_2<f_2,f_3>}&=(0.0,0.0,0.0,0.0,0.0,0.0,0.0,1.0,0.2)\end{aligned}$$

For sample x_3, given the feature combination reference sample $P=\{<0.5,0.2>,<0.5,0.4>,<0.2,0.4>\}$, the results obtained using the aforementioned multi-distance metric granularization methods are:

$$G_{x_3<f_1,f_2>} = (0.8, 0.63246, 0.4, 0.10909, 0.6, 0.76667, 0.36364, 0.89091, 0.51)$$
$$G_{x_3<f_1,f_3>} = (0.29999, 0.22361, 0.04999, 0.00148, 0.19999, 0.27778, 0.14286, 0.99852, 0.55)$$
$$G_{x_3<f_2,f_3>} = (0.7, 0.60828, 0.37, 0.14672, 0.6, 0.71111, 0.36842, 0.85328, 0.36)$$

Thus, the results of multi-distance metric granularization for samples x_1, x_2, x_3 under the reference sample $P = (0.5, 0.2, 0.4)$, formed by multi-feature combinations into a coordinate plane system are:

$$\begin{aligned} G_{x_1} &= (G_{x_1<f_1,f_2>}, G_{x_1<f_1,f_3>}, G_{x_1<f_2,f_3>})^T \\ &= \{(0.5, 0.36056, 0.13, 0.20384, 0.3, 0.67857, 0.33333, 0.79616, 0.25), (0.39999, 0.28284, 0.07999, 0.09204, 0.2, 0.44999, 0.22222, 0.90796, 0.39), (0.49999, 0.36056, 0.12999, 0.02658, 0.3, 0.62857, 0.29412, 0.97342, 0.33999))\}^T \end{aligned}$$

$$\begin{aligned} G_{x_2} &= (G_{x_2<f_1,f_2>}, G_{x_2<f_1,f_3>}, G_{x_2<f_2,f_3>})^T \\ &= \{(0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 1.0, 0.29), (0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 1.0, 0.41), (0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 1.0, 0.2)\}^T \end{aligned}$$

$$\begin{aligned} G_{x_3} &= (G_{x_3<f_1,f_2>}, G_{x_3<f_1,f_3>}, G_{x_3<f_2,f_3>})^T \\ &= \{(0.8, 0.63246, 0.4, 0.10909, 0.6, 0.76667, 0.36364, 0.89091, 0.51), (0.29999, 0.22361, 0.04999, 0.00148, 0.19999, 0.27778, 0.14286, 0.99852, 0.55), (0.7, 0.60828, 0.37, 0.14672, 0.6, 0.71111, 0.36842, 0.85328, 0.36)\}^T \end{aligned}$$

The unfolding of samples x_1, x_2, x_3 based on the number of feature combinations on the decision feature L is as follows:

$$GL_{x_1} = (0, 0, 0)$$
$$GL_{x_2} = (1, 1, 1)$$
$$GL_{x_3} = (0, 0, 0)$$

3.2 Multi-distance metrics for granular vectors

The concept of multi-distance metrics for granular vectors is introduced in relation to the definition of multiple distance metrics applied to granules. These metrics are designed to capture, and in some respects, granulate, the similarities and differences between distinct granules. By employing various distance metric approaches, such as Euclidean distance, Manhattan distance, and Chebyshev distance, the relationships among granular vectors can be analyzed in a comprehensive manner.

Definition 10 Suppose that given two m-dimensional granular vectors $G = \{g_1, g_2, \ldots, g_m\}^T$ and $H = \{h_1, h_2, \ldots, h_n\}^T$, and a single-distance granularization method $d \in D$, where each granule has n granular cores, the result of applying the granularization method d between granular vector G and H is:

$$d(G, H) = \sum_{i=0}^{m} \sum_{j=0}^{n} \sum_{k=j+1}^{n} v(g_{i_{<r_j,r_k>}}, h_{i_{<r_j,r_k>}}) \tag{25}$$

where r_j and r_k are the granular cores at the corresponding position of the vector elements, and v is the distance calculated between G and H based on the granularization method d.

Definition 11 Suppose that given two m-dimensional granular vectors $G = \{g_1, g_2, \ldots, g_m\}^T$ and $H = \{h_1, h_2, \ldots, h_n\}^T$, where each granule has n granular cores, several multi-distance metric methods for granular vectors are given as follows:

Manhattan Distance:

$$D_h(G, H) = \sum_{i=1}^{m} \sum_{j=1}^{n} \sum_{k=j+1}^{n} (\left|g_{i_{r_j}} - h_{i_{r_j}}\right| + \left|g_{i_{r_k}} - h_{i_{r_k}}\right|), \tag{26}$$

Euclidean distance:

$$D_e(G, H) = \sum_{i=1}^{m} \sum_{j=1}^{n} \sum_{k=j+1}^{n} \sqrt{\left(g_{i_{r_j}} - h_{i_{r_j}}\right)^2 +} \sqrt{\left(g_{i_{r_k}} - h_{i_{r_k}}\right)^2}, \tag{27}$$

Squared Euclidean distance:

$$D_q(G, H) = \sum_{i=1}^{m} \sum_{j=1}^{n} \sum_{k=j+1}^{n} \left((g_{i_{r_j}} - h_{i_{r_j}})^2 + (g_{i_{r_k}} - h_{i_{r_k}})^2\right), \tag{28}$$

Cosine distance:

$$D_c(G, H) = 1 - \frac{\sum_{i=1}^{m} \sum_{j=1}^{n} \sum_{k=j+1}^{n} (g_{i_{r_j}} {}^* h_{i_{r_j}} + g_{i_{r_k}} {}^* h_{i_{r_k}})}{\sqrt{\sum_{i=1}^{m} \sum_{j=1}^{n} \left(g_{i_{r_j}} {}^* g_{i_{r_j}} + h_{i_{r_j}} {}^* h_{i_{r_j}}\right)}} * \frac{1}{\sqrt{\sum_{i=1}^{m} \sum_{j=1}^{n} \sum_{k=j+1}^{n} (g_{i_{r_k}} {}^* g_{i_{r_k}} + h_{i_{r_k}} {}^* h_{i_{r_k}})}}, \tag{29}$$

Chebyshev Distance:

$$D_{cb}(G,H) = \max(\sum_{i=1}^{m}\sum_{j=1}^{n}\left|g_{i_{r_j}} - h_{i_{r_j}}\right|, \sum_{i=1}^{m}\sum_{j=1}^{n}\sum_{k=j+1}^{n}\left|g_{i_{r_k}} - h_{i_{r_k}}\right|), \tag{30}$$

Canberra Distance

$$D_{ca}(G,H) = \sum_{i=1}^{m}\sum_{j=1}^{n}\frac{\left|g_{i_{r_j}} - h_{i_{r_j}}\right|}{\left|g_{i_{r_j}} + h_{i_{r_j}}\right|} + \sum_{i=1}^{m}\sum_{j=1}^{n}\sum_{k=j+1}^{n}\frac{\left|g_{i_{r_k}} - h_{i_{r_k}}\right|}{\left|g_{i_{r_k}} + h_{i_{r_k}}\right|} \tag{31}$$

Bray Curtis Distance:

$$D_b(G,H) = \frac{\sum_{i=1}^{m}\sum_{j=1}^{n}\left|g_{ir_j} - h_{ir_j}\right| + \sum_{i=1}^{m}\sum_{j=1}^{n}\sum_{k=j+1}^{n}\left|g_{ir_k} {}^{*} h_{ir_k}\right|}{\sum_{i=1}^{m}\sum_{j=1}^{n}\sum_{k=j+1}^{n}(|g_{ir_j}| + |g_{ir_k}| + |h_{ir_j}| + |h_{ir_k}|)}, \tag{32}$$

Ochiai distance:

$$D_o(G,H) = \frac{\sum_{i=1}^{m}\sum_{j=1}^{n}\sum_{k=j+1}^{n}(g_{i_{r_j}} * h_{i_{r_j}} + g_{i_{r_k}} * h_{i_{r_k}})}{\sqrt{\sum_{i=1}^{m}\sum_{j=1}^{n}\sum_{k=j+1}^{n}(g_{i_{r_j}} * g_{i_{r_j}} + g_{i_{r_k}} * g_{i_{r_k}})}} * \frac{1}{\sqrt{\sum_{i=1}^{m}\sum_{j=1}^{n}\sum_{k=j+1}^{n}(h_{i_{r_j}} * h_{i_{r_j}} + h_{i_{r_k}} * h_{i_{r_k}})}}, \tag{33}$$

Product distance:

$$D_p(G,H) = \sum_{i=1}^{m}\sum_{j=1}^{n}\sum_{k=j+1}^{n}(g_{i_{r_j}} * h_{i_{r_j}} + g_{i_{r_k}} * h_{i_{r_k}}), \tag{34}$$

where $g_{i_{r_j}}$ represents the $j-th$ granular core of the granular vector g_i.
This multi-distance metric granulation not only enriches the data representation but also enhances the visibility of minority class samples in the transformed feature space. Since different distance metrics emphasize different local structures, the minority instances are more likely to be preserved as distinct patterns during feature transformation, thereby implicitly improving class imbalance robustness.

4 The granlar XGBoost classification algorithm

4.1 Principles of granular XGBoost classification

Based on the XGBoost algorithm, we propose the integration of granular computing to develop Granular XGBoost (GXGBoost). To address the issue of overfitting in small-sample scenarios, the granularization process transforms original features into multi-level representations using multiple distance metrics. These representations encode local structure and suppress sample-specific noise, enabling the XGBoost classifier to generalize better and make more stable split decisions during tree construction. To further enhance learning stability, the algorithm applies min-max normalization to rescale all features to a uniform range within [0,1]:

$$X_{norm} = \frac{X - X_{min}}{X_{max} - X_{min}} \tag{35}$$

This transformation preserves the original data distribution while maintaining numerical stability for distance computations.

The algorithm employs a completely random sampling strategy to select reference samples. Regardless of whether the data set is balanced or not, the number of reference samples *S* should be equal to the number of classes $K(S = K)$, in order to maximize the probability of class coverage while suppressing noise introduction.

Theorem 1 *Consider a dataset containing K classes with N total samples, where class i has N_i samples (allowing for imbalanced cases where $N_i \ll N$). When randomly selecting S reference samples:The probability of covering at least one sample from class i is:*

$$P_{cover}(i) = 1 - \left(1 - \frac{N_i}{N}\right)^S \tag{36}$$

When $S = K$, $\sum_{i=1}^{K} P_{cover}(i)$ attains a local maximum (guaranteed by the inclusion-exclusion principle).

Proof of Theorem 1 1. ***When*** $S < K$:

- For any class i (particularly minority classes where $N_i \ll N$), the coverage probability $P_{\text{cover}}(i) = 1 - \left(1 - \frac{N_i}{N}\right)^S$ decreases exponentially with *S*.
- For balanced data ($N_i \approx N/K$), $P_{\text{cover}}(i)$ remains low, leading to underfitting.

- For imbalanced data, $P_{\text{cover}}(i) \approx 0$ for minority classes, preventing the model from learning their distributions.

2. ***When*** $S > K$:

- Redundant reference samples primarily come from the majority class (since $N_1 \gg N_i$).
- Noise variance $\epsilon_{\text{noise}} \propto (S - K) \cdot \frac{N_1}{N}$ grows linearly with *S*, reducing generalization capability.

3. ***When*** $S = K$:

- For balanced data: $P_{\text{cover}}(i) \approx 1 - e^{-1} \approx 63\%$ (uniform coverage across classes).
- For imbalanced data: Multi-distance metrics (e.g., Manhattan distance's sensitivity to sparse features) compensate for low minority class coverage, while ensemble learning further mitigates bias.$\square$

The GXGBoost classification algorithm primarily employs a method rooted in multi-distance metric granulation. The core concept first extends the dimensionality of the initial dataset with *h* features by pairwise combinations, generating $\frac{h*(h-1)}{2}$ new features. Each feature combination is then granularized using *L* distance metrics (e.g., Manhattan, Cosine), producing multi-granularity representations. These representations are directly concatenated into unified vectors that serve as input to *L* parallel XGBoost classifiers, each corresponding to a specific distance metric. In GXGBoost, the tree-building process operates on granular vectors rather than raw features. These vectors aggregate multi-distance-based perspectives, providing stable and generalized criteria for split evaluation. Consequently, each decision tree is built upon abstracted patterns rather than raw, potentially noisy data, thus enhancing robustness and reducing variance. To mitigate potential noise introduced by direct concatenation, each XGBoost classifier incorporates regularization terms $\Omega(f_t)$ during tree construction. During training, each classifier iteratively constructs decision trees by optimizing splits that maximize class coverage probabilities, with the process terminating when the empirical risk stabilizes, thus preventing overfitting.

At the $t-th$ tree generated by this GXGBoost after training, the objective function of this decision tree is:

$$Obj^{(t)} = \sum_{i=1}^{n} l\Big(GL_{p(i,j)}, \widehat{GL}_{p(i,j)}^{(t)}\Big) + \Omega(f_t) + constant, \tag{37}$$

$$l\Big(GL_{p(i,j)}, \widehat{GL}_{p(i,j)}^{(t)}\Big) = |GL_{p(i,j)} - \widehat{GL}_{p(i,j)}^{(t)}|, \tag{38}$$

$\Omega(f_t)$ represents the regularization term to control the complexity of the model, *i.e.*, to control the structural risk of the current tree. The principle of the GXGBoost classification algorithm is shown in Fig. 1.

4.2 The granular XGBoost classification algorithm

In the previous section, the principle of GXGBoost classification was explained, and this section will elaborate on the GXGBoost classification Algorithm 1.

Algorithm 1 Granular XGBoost (GXGBoost) Classification Algorithm

Input: Multi-granular system $M = (X, F, P, L, D)$ where:

- $X = \{x_1, x_2, \ldots, x_n\}$: Sample set
- $F = \{f_1, ..., f_m\}$: Feature set
- $P = \{p_1, p_2, \ldots, p_q\}$: Reference samples
- $L \in \{0, 1\}^n$: Decision labels
- $D = \{d_1, ..., d_k\}$: Distance metrics

Output: Predicted labels $\hat{L} \in \{0, 1\}^n$

1. Data Normalization
$X \leftarrow \text{MinMaxScaler}(X)$
2. Reference Aggregation
$P \leftarrow \text{Complete Random Sample}(X, q)$
3. Feature Pair Expansion
$EX \leftarrow \{\langle x_i, (f_a, f_b)_{a<b}\rangle \mid x_i \in X\}$
$EP \leftarrow \{\langle P_i, (f_a, f_b)_{a<b}\rangle \mid p_i \in P\}$
4. Granular Tensor Construction
$GX \leftarrow \emptyset$
for EX_i in EX **do**
  for EP_i in EP **do**
    for d_i in D **do**
      $GX \leftarrow \{GX, d_i(EX_i, EP_i)\}$
    end for
  end for
end for
5. Partitioning
$\{GX_l \leftarrow \text{split}(GX, d_i)$
6. Parallel Training
Decision Tree $\leftarrow \{\text{XGBoost}(GX_l) \mid l = 1..k\}$
7. Ensemble Prediction
$\hat{L} \leftarrow$ majority_vote(Decision Tree)

4.3 Complexity analysis

Let *m* be the number of original features, *q* the number of reference samples, *d* the number of distance metrics, and *n* the number of data samples. The computational complexity of the GXGBoost algorithm can be analyzed in three main stages:

- Feature Combination Phase: The total number of pairwise feature combinations is $\frac{m*(m-1)}{2}$, leading to a time complexity of $\mathcal{O}(m^2 \cdot n)$.
- Multi-Distance Granulation Phase: Each sample is compared with q reference samples across d distance

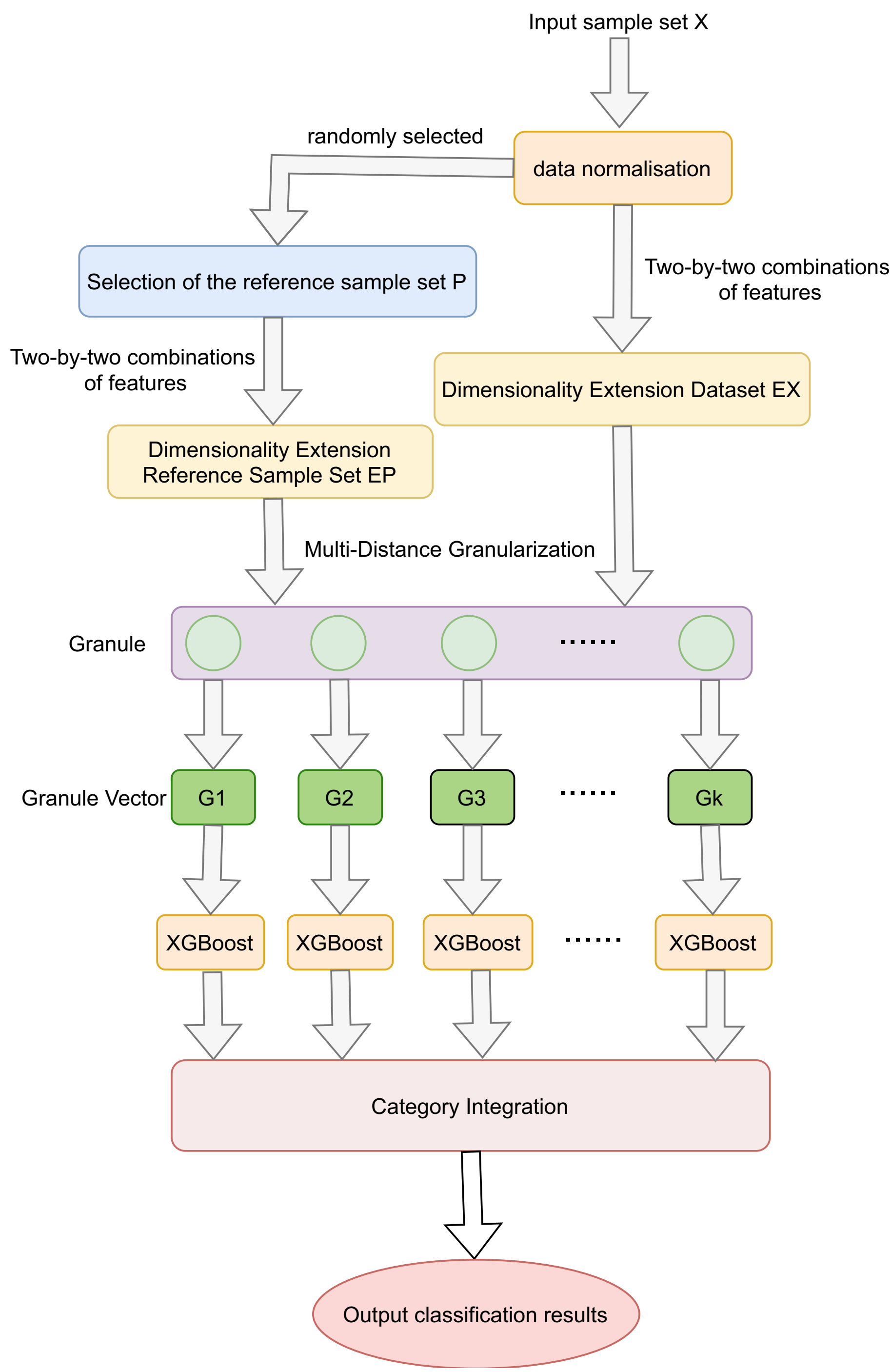

Fig. 1 The granular XGBoost classification algorithm

metrics. For each feature combination, the distance is computed, resulting in complexity $\mathcal{O}(d \cdot q \cdot m^2 \cdot n)$.

- XGBoost Training Phase: The number of input features is increased to $\mathcal{O}(d \cdot m^2)$, and assuming standard tree-building complexity, the time complexity per tree becomes $\mathcal{O}(d \cdot m^2 \cdot nlogn)$.

Thus, the total time complexity of GXGBoost is:

$$\mathcal{O}\left(\underbrace{m^2 \cdot n}_{\text{Feature Combination}} + \underbrace{q \cdot d \cdot m^2 \cdot n}_{\text{Granulation}} + \underbrace{d \cdot m^2 \cdot n \log n}\right)$$

The space complexity is dominated by the granular feature matrix, with complexity:

$$\mathcal{O}\left(d \cdot q \cdot m^2 \cdot n\right)$$

Table 2 Overview of the datasets

Datasets	Number of Samples	Number of Features	Number of Categories
Iris	150	4	3
Wine Quality	4898	11	3
Wine	178	13	3
Haberman's Survival	306	3	2
Heart Disease	303	13	2
Breast Cancer Coimbra	116	9	2
Titanic	91	8	2
Pima Indians	768	8	2
Abalone	4177	8	3
Balance Scale	625	4	3
Breast Tissue	106	9	6
Sonar	208	60	2
Energy Efficiency	768	8	4
HCV	615	12	5
Indian Liver Patient Dataset	583	10	2
Ionosphere	351	34	2
Parkinsons	197	22	2
Seeds	210	7	3
Teaching Assistant Evaluation	151	5	3
Lymphography	148	19	4
Vertebral Column	310	6	2
Wholesale customers	440	7	3
Blood Transfusion Service Center	748	4	2

5 Experimental analysis

In the context of the GXGBoost classification algorithm, selecting an appropriate number of reference samples and a suitable distance metric is crucial, as this decision depends on the composition of the dataset. Datasets were selected based on three criteria: (1) Sample size <10,000 to focus on small-data regimes; (2) Diversity in class distribution, including both imbalanced (class ratio $\geq$1:3) and balanced datasets (class ratio $\leq$1:1.5); (3) Feature diversity, covering low-dimensional ($\leq$10 features) and optionally high-dimensional scenarios.For dataset selection criteria condition 2, a partially balanced dataset was selected to provide a baseline comparison and to validate the generality of the model on balanced data.For this study, 23 frequently used datasets from UCI and Kaggle were selected. Each data set exhibits distinct characteristics and class distributions; for example, the Titanic dataset originally contained 11 characteristics, and this research focuses on a selection of the most commonly used features, as described in Table 2.

To obtain more accurate results, the XGBoost and GXGBoost classification algorithms were trained on six datasets for 50 rounds, using accuracy as the evaluation metric. As illustrated in Fig. 2, although the accuracy of the XGBoost algorithm was slightly higher than that of the GXGBoost algorithm in certain instances, the GXGBoost classification algorithm exhibited significantly superior accuracy in the majority of cases. This finding indicates that the observed results are not coincidental.

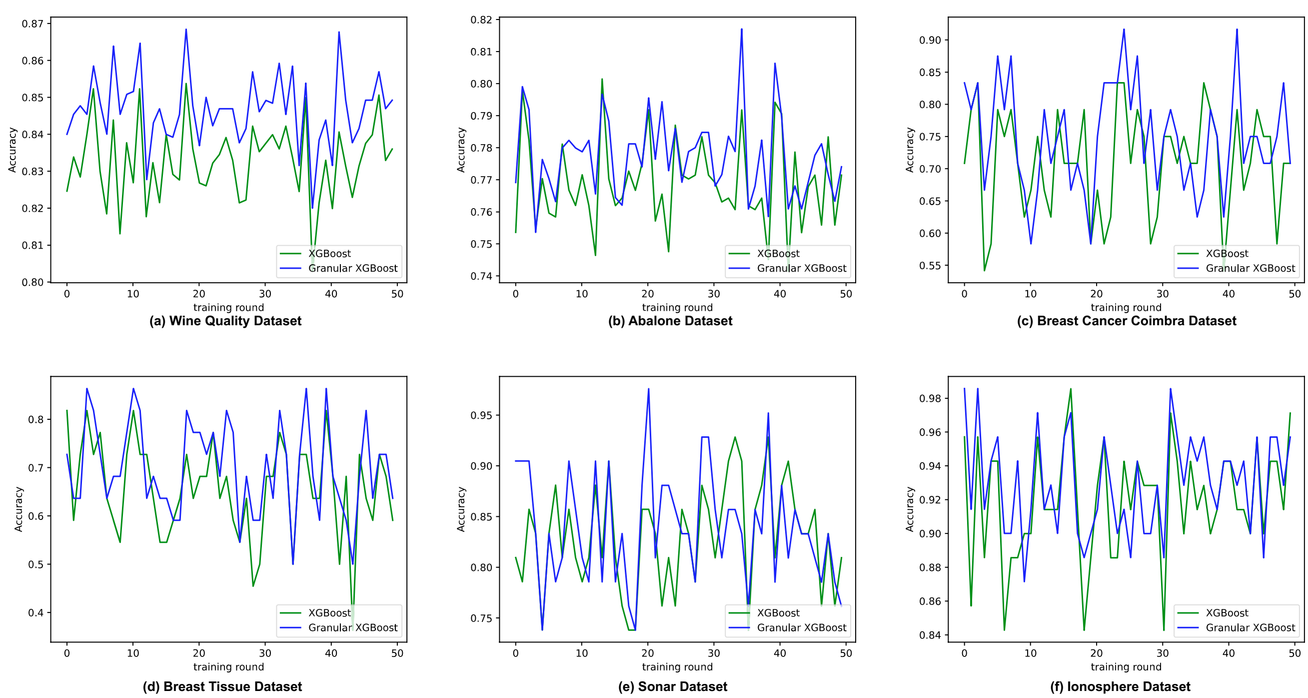

Fig. 2 Comparison of multi-rounds of training

In this experiment, multiple independent training sessions were conducted in six datasets: the Wine Quality dataset and the Abalone dataset, which contain the largest number of samples; the Breast Cancer Coimbra dataset and the Breast Tissue dataset, which have the smallest number of samples; and the Sonar dataset and the Ionosphere dataset, which feature the highest number of attributes. The classification accuracy was recorded for each training session. During more than 50 rounds of training, the GXGBoost algorithm consistently demonstrated stable and excellent performance, underscoring its advantages in managing both high-dimensional complex data and small-sample data. In contrast, while XGBoost performed well in certain rounds, it was generally less effective than GXGBoost overall.

5.1 Impact of the reference samples

The number of reference samples plays a critical role in influencing both the granularity of the model and the performance of the resulting classifier. To start with, choosing a balanced quantity of reference samples is essential for accurately reflecting the distribution characteristics of the dataset, thereby improving the model's generalization skills. Lack of adequate reference samples may result in overfitting, causing the model to become overly tailored to particular instances, which ultimately undermines its generalization capabilities. In such cases, the model might focus on noise and outliers within the data rather than recognizing broader patterns, potentially leading to poor performance on validation datasets. However, having too many reference samples can increase computational requirements, resulting in longer training times and inefficient use of resources. Hence, it is vital to establish an optimal number of reference samples. This ideal count can improve computational efficiency without sacrificing the precision of the model. Such a strategy not only improves the model's ability to effectively process and predict new data but also shortens the training duration. As a result, carefully selecting an appropriate number of reference samples is fundamental to enhancing model performance, ensuring computational efficacy, and achieving balanced generalization capacity.

To better understand the impact of reference sample quantity on the GXGBoost classification algorithm, a series of experiments were conducted using a dataset with varying numbers of reference samples. In these experiments, a consistent distance metric was employed for granulation, with accuracy as evaluation criterion. A comparison was made between the XGBoost classification algorithm and the enhanced multi-distance granular XGBoost classification algorithm (GXGBoost). The results of the experiments, which used different numbers of reference samples, are illustrated in Fig. 3. This experiment verifies the optimality of the number of reference samples $S = K$ (number of categories). The experimental results are highly consistent with the theoretical predictions and are analyzed as follows.

Figure 3(a) illustrates that, in the Iris dataset, when the number of reference samples is set to 1, the accuracy of the GXGBoost classifier is slightly higher than that of the XGBoost classifier. As the number of reference samples increases to 3, the Iris dataset achieves its peak accuracy. However, with further increases in the number of reference samples, the accuracy begins to fluctuate. For instance, accuracy decreases when the number of reference samples reaches 4, and after 5, it falls below that of the original dataset. Similarly, Fig. 3(b) indicates that, for the Wine dataset, the accuracy of the GXGBoost classifier exhibits significant fluctuations, peaking at a reference sample number of 3. Beyond this point, accuracy declines sharply, reaching a trough at a reference sample number of 6. Although there is a slight rebound thereafter, it never surpasses the performance of the XGBoost classifier. In the Haberman's Survival dataset (Fig. 3(c)), the accuracy of GXGBoost consistently exceeds that of XGBoost, with optimal performance observed at a reference sample number of 2. In the Heart Disease dataset (Fig. 3(d)), GXGBoost achieves its highest accuracy at a reference sample number of 2, followed by a decrease at 3, and subsequently, its accuracy is significantly lower than that of the XGBoost classifier when the reference sample size is 4 or greater. Lastly, in the Breast Cancer Coimbra dataset (Fig. 3(e)), the highest accuracy is attained at a reference sample number of 2, after which there are fluctuations that generally reflect a downward trend as the reference sample number exceeds 2. In the Titanic dataset (Fig. 3(f)), the performance of the GXGBoost classifier is only slightly superior to that of XGBoost when the number of reference samples is 2; otherwise, its performance is generally poor. In the Breast Tissue dataset (Fig. 3(g)), the accuracy across various numbers of reference samples consistently exceeds that of the XGBoost classifier, achieving the highest accuracy with 6 reference samples. For the Pima Indians dataset (Fig. 3(h)), the GXGBoost classifier outperforms XGBoost with 2 and 4 reference samples, attaining its peak accuracy at 2 samples, while the performance in other cases is lower than that of XGBoost. Lastly, in the Ionosphere dataset (Fig. 3(i)), the GXGBoost classifier consistently demonstrates higher accuracy than XGBoost, regardless of the number of reference samples, achieving its maximum accuracy with 4 reference samples.

The combined dataset indicates that the Iris and Wine datasets each contain 3 categories, the Breast Tissue dataset comprises 6 categories, the Ionosphere dataset includes 4 categories, while the remaining datasets have 2 categories each. The results demonstrate that model performance peaks when the number of reference samples (S) matches the

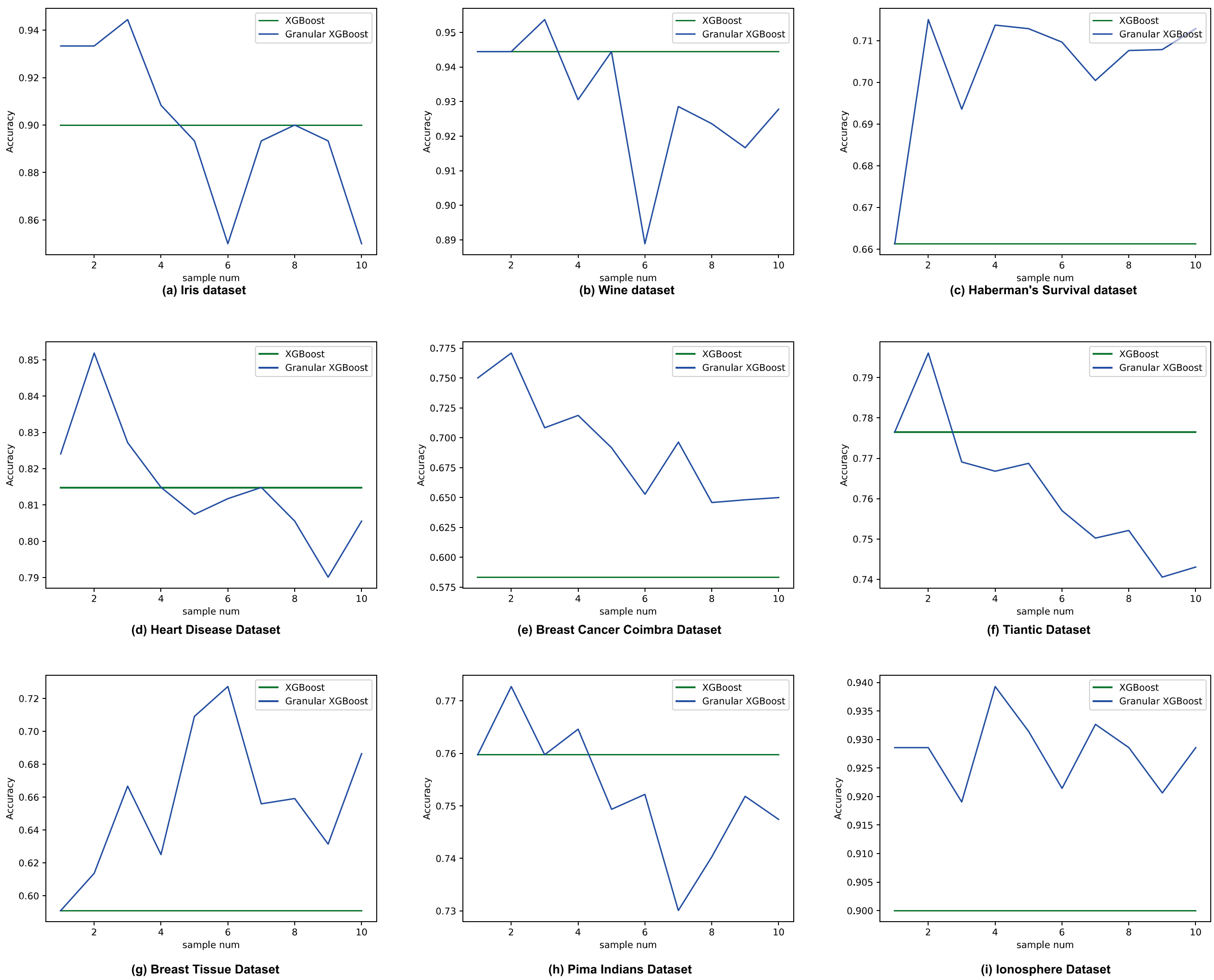

Fig. 3 Classification accuracy of datasets with different numbers of reference samples

dataset's inherent category count (K). However, exceeding this critical threshold (S > K) introduces redundant details that degrade generalization capability, likely due to noise accumulation. These findings conclusively validate the theoretical premise: optimal classification accuracy is achieved under the condition S = K, where maximal class coverage probability and minimal noise variance are balanced.

5.2 Impact of distance metric methods

In classification tasks, the selection of a suitable distance metric plays a crucial role in determining the overall effectiveness of a model's performance. Relying on a single distance metric often falls short of encapsulating the full spectrum of diversity and complexity inherent in the data. On the other hand, the introduction of a combined approach that integrates multiple distance metric methods has the potential to harness the advantages of various techniques. This amalgamation results in a more thorough and nuanced representation of features, which can lead to improved classification outcomes. To enhance the algorithm's effectiveness and boost the performance of the model, this section explores the impact of using different numbers of distance metric methods on the GXGBoost classifier.

Experiments were conducted using various combinations of distance metric methods across each dataset. Given that a single distance metric may not adequately capture the complexity and diversity of the data, Manhattan distance was selected as the initial distance metric for the experiments, following this, between 1 and 8 additional distance metrics were employed. The resulting changes in classification accuracy for XGBoost and GXGBoost across each dataset, utilizing the optimal number of reference samples, are illustrated in Fig. 4.

In this study, we evaluated the impact of nine distance metric methods on multiple datasets. These methods include

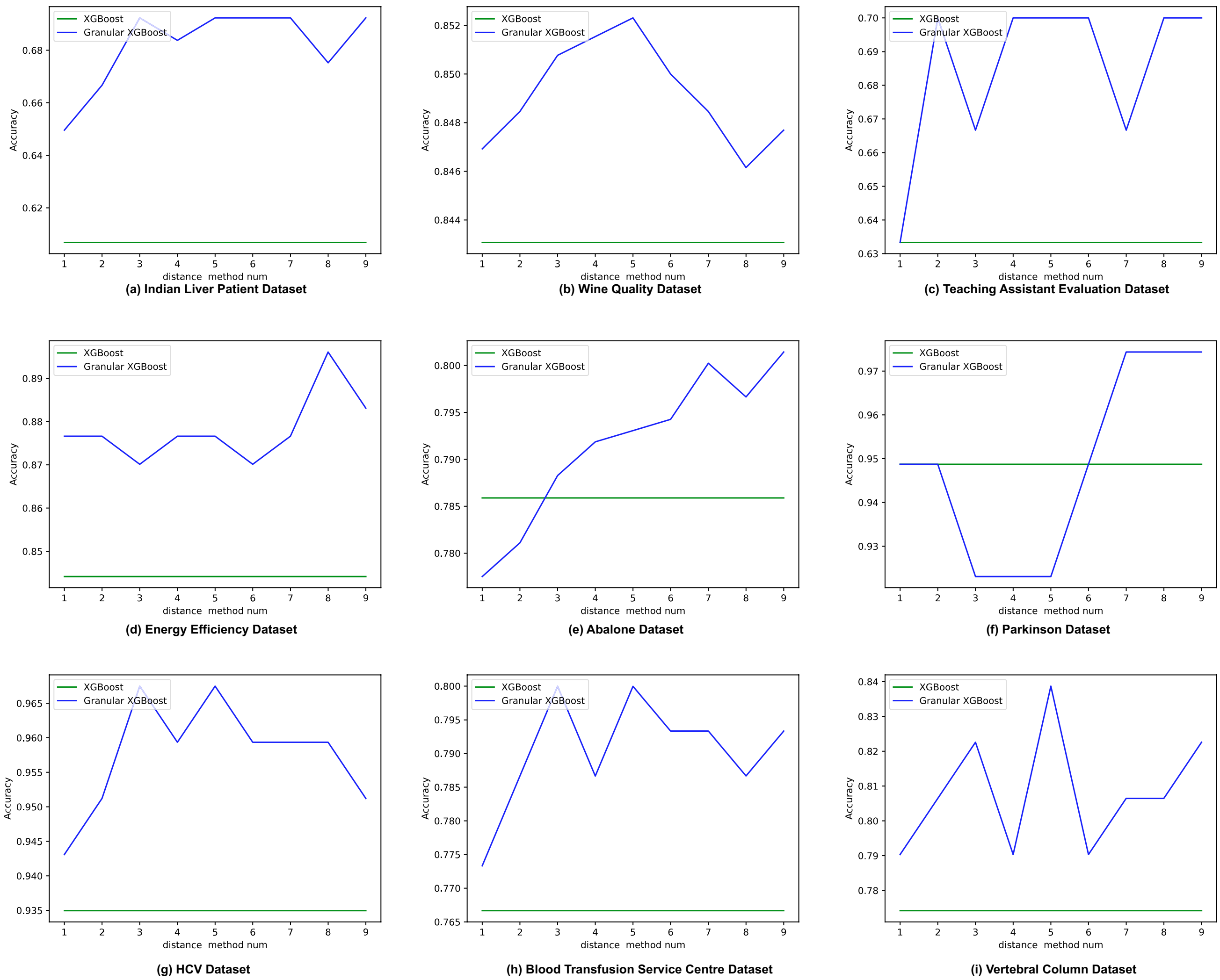

Fig. 4 Classification accuracy of the dataset with different numbers of distance metric methods

Manhattan distance, Euclidean distance, Squared Euclidean distance, Cosine distance, Chebyshev distance, Canberra distance, Bray Curtis distance, Ochiai distance, and Product distance. By analyzing the results presented in Fig. 4, we can draw the following conclusions: From Fig. 4(a), the Indian Liver Patient Dataset demonstrates a significant increase in classification accuracy when utilizing the first three distance metrics (Manhattan Distance, Euclidean Distance, and Squared Euclidean Distance). However, accuracy decreased with the addition of the fourth method (Cosine distance), followed by an increase with the fifth method (Chebyshev distance). The accuracy remained stable when the sixth and seventh methods (Canberra distance and Bray Curtis distance) were applied, but decreased again with the introduction of the eighth method (Ochiai distance). Finally, the accuracy increased once more when employing the ninth method (Product distance).

In Fig. 4(b), the Wine Quality dataset demonstrates a substantial increase in accuracy when employing distance metrics from the first to fifth methods (Manhattan distance, Euclidean distance, Squared Euclidean distance, Cosine distance, and Chebyshev distance). Conversely, there is an almost linear decrease in accuracy with the addition of the sixth to eighth methods (Canberra distance, Bray Curtis distance, and Ochiai distance). Notably, classification accuracy improves again with the introduction of the ninth method (Product distance). Figure 4(c) illustrates that the Teaching Assistant Evaluation dataset experiences a significant enhancement in accuracy when utilizing the second (Euclidean distance), fourth (Cosine distance), and eighth (Ochiai distance) methods.

Figure 4(d) demonstrates that the Energy Efficiency dataset exhibits improved accuracy compared to the XGBoost classifier with the inclusion of the first method (Manhattan distance). This is followed by enhancements in accuracy with the addition of the fourth (Cosine distance), seventh (Bray Curtis distance), and eighth (Ochiai distance) methods, all of which surpass the performance of the previous distance metrics. In Fig. 4(e), it is observed that the Abalone dataset performs with lower accuracy than the XGBoost classifier when incorporating the first (Manhattan Distance) and second (Euclidean Distance) methods. However, there is a notable increase in classification accuracy with the addition of the third to seventh methods (Squared Euclidean Distance, Cosine Distance, Chebyshev Distance, Canberra Distance, and Bray Curtis distance), followed by a slight decrease in accuracy with the eighth method (Ochiai distance) and a subsequent increase with the ninth method (Product distance). Figure 4(f) reveals that the Parkinson dataset shows no change in accuracy with the first method (Manhattan distance). A significant decrease in accuracy is observed after adding the second (Euclidean distance) and third (Squared Euclidean distance) methods. The accuracy remains unchanged after incorporating the fourth (Cosine distance) and fifth (Chebyshev distance) methods, but there is a significant improvement following the inclusion of the sixth (Canberra distance) and seventh (Bray Curtis distance) methods. Finally, Fig. 4(g) indicates that the classification accuracy of the HCV dataset improves with the addition of the first (Manhattan distance), second (Euclidean distance), third (Squared Euclidean distance), and fifth (Chebyshev distance) methods.

In summary, different datasets exhibited varying responses to the various distance metric methods. The Indian Liver Patient Dataset was most effectively analyzed using Manhattan distance, Euclidean distance, Squared Euclidean distance, Chebyshev distance, and Product distance. The Wine Quality dataset showed optimal results with Manhattan distance, Euclidean distance, Squared Euclidean distance, Cosine distance, Chebyshev distance, and Product distance. The Teaching Assistant Evaluation dataset excelled with Euclidean distance, Cosine distance, and Ochiai distance. The Energy Efficiency dataset performed well with Manhattan distance, Cosine distance, Bray-Curtis distance, and Ochiai distance. The Abalone dataset demonstrated superior performance with Squared Euclidean distance, Cosine distance, Chebyshev distance, Canberra distance, Bray-Curtis distance, and Product distance. The Parkinson dataset was best suited for Canberra distance and Bray-Curtis distance. The HCV dataset yielded better results with Manhattan distance, Euclidean distance, Squared Euclidean distance, and Chebyshev distance. The Blood Transfusion Service Center dataset was also suitable for Manhattan distance, Euclidean distance, Squared Euclidean distance, Chebyshev distance, and Product distance. Finally, the Vertebral Column dataset achieved good performance with Manhattan distance, Euclidean distance, Squared Euclidean distance, Chebyshev distance, Bray-Curtis distance, and Product distance.

The Indian Liver Patient Dataset, the Wine Quality Dataset, and the Vertebral Column Dataset belong to continuous numerical feature datasets. The Teaching Assistant Evaluation dataset and the Energy Efficiency dataset are sparse datasets, the Abalone dataset and Parkinson dataset are datasets with large differences in feature values, and both the HCV dataset and Blood Transfusion Service Center dataset consist of continuous numerical and discrete features.

Based on the classification accuracy across different distance metric and the shared characteristics of these datasets, the following conclusions can be drawn:

- Manhattan, Euclidean, Squared Euclidean, Chebyshev, and Product distances are well suited for datasets with continuous numerical features.
- Cosine and Ochiai distances perform best on sparse datasets.
- Canberra and Bray-Curtis distances are appropriate for datasets with large disparities in feature magnitudes.
- For datasets combining continuous and discrete features, Manhattan, Euclidean, Squared Euclidean, and Chebyshev distances are effective.

The GXGBoost classification algorithm needs to improve the accuracy of the classifier and optimize the performance of the model by choosing the appropriate distance metrics according to the characteristics of the dataset.

5.3 Algorithmic complexity

To validate the theoretical analysis in Section 4.3, we conducted experiments on synthetic datasets with varying sample sizes (n), feature counts (m), number of reference samples (q), and distance metrics (d). The tests were performed on a standard desktop environment (Intel i7 CPU, 16GB RAM), and results were averaged over 5 independent runs. Tables 3 and 4 report the time and space consumption for different configurations.

Table 3 Time complexity results

n	m	k	d	$feature\_comb\_time$	$granulation\_time$	$training\_time$	$total\_time$	$XGBoost\_time$
100	3	2	3	0.0023057	0.0753951	2.18418	2.26188	0.0602414
500	10	5	5	0.0348061	2.80316	1.27717	4.11513	0.0867287
1000	30	5	7	0.261434	28.498	14.2095	42.9689	0.1491993
5000	50	10	9	0.671665	178.04	68.4247	247.136	0.1976011

Table 4 Space complexity results

n	m	k	d	$feature\_comb\_mem$	$granulation\_mem$	$train\_mem$	$total\_mem$	$XGBoost\_mem$
100	3	2	3	31768	301117	262632	595517	150891
500	10	5	5	3509100	16620805	333526	20463431	217867
1000	30	5	7	69712372	314042149	384028	384138549	241873
5000	50	10	9	980329460	8828444645	497428	9809271533	349168

As shown in Table 3, the total running time increases non-linearly with respect to n, m, and d, aligning with the theoretical complexity of $\mathcal{O}(d \cdot q \cdot m^2 \cdot n)$. Notably, the time required by the granulation phase dominates the total time in large configurations, such as when n=5000 and m=50.

Similarly, Table 4 demonstrates that memory usage scales steeply, especially due to the need to store granulated feature matrices. Compared to traditional XGBoost, whose training memory footprint remains relatively low, GXGBoost consumes significantly more memory, particularly in the granulation stage, which confirms the theoretical analysis of the complexity of the space.

These results highlight the trade-off between the richer feature representation in GXGBoost and the increased computational cost. However, the improved classification performance (as shown in Section 5.2) justifies this cost in many small-sample and high-complexity scenarios.

Despite the increased computational demands introduced by the granulation module, the performance improvements demonstrated in Section 5.2 clearly indicate that the trade-off is favorable. In scenarios with small sample sizes or noisy data, the additional cost is justified by the significantly improved robustness and precision of the model, confirming the practical value of the proposed GXGBoost framework.

5.4 Interpretability analysis and feature importance comparison

To assess the interpretability of the proposed Granular XGBoost (GXGBoost) model and compare it with the traditional XGBoost in terms of importance of features, we conducted a comprehensive analysis from three perspectives: importance of original features in XGBoost, importance of granular features in GXGBoost, and global explanation using SHAP (SHapley Additive exPlanations).

The left part of Fig. 5 presents the importance of the feature of the standard XGBoost model based on the F score. The most important features are f0 through f3, indicating that the model relies heavily on early original features to make decisions.

Right part of Fig. 5 shows the importance of the feature in the GXGBoost model, where the most influential features are combinations of original features, such as f5_f7, f5_f8, f5_f9, f5_f10 and f5_f11. This indicates that GXGBoost prioritizes interactions between certain features, particularly those involving f5. Frequency analysis of the feature combinations reveals that f7, f10, and f11 each appear in six combinations, followed closely by f5, f6, f8, and f9, each appearing five times. This suggests that GXGBoost emphasizes inter-feature relationships involving the latter part of the original feature set.

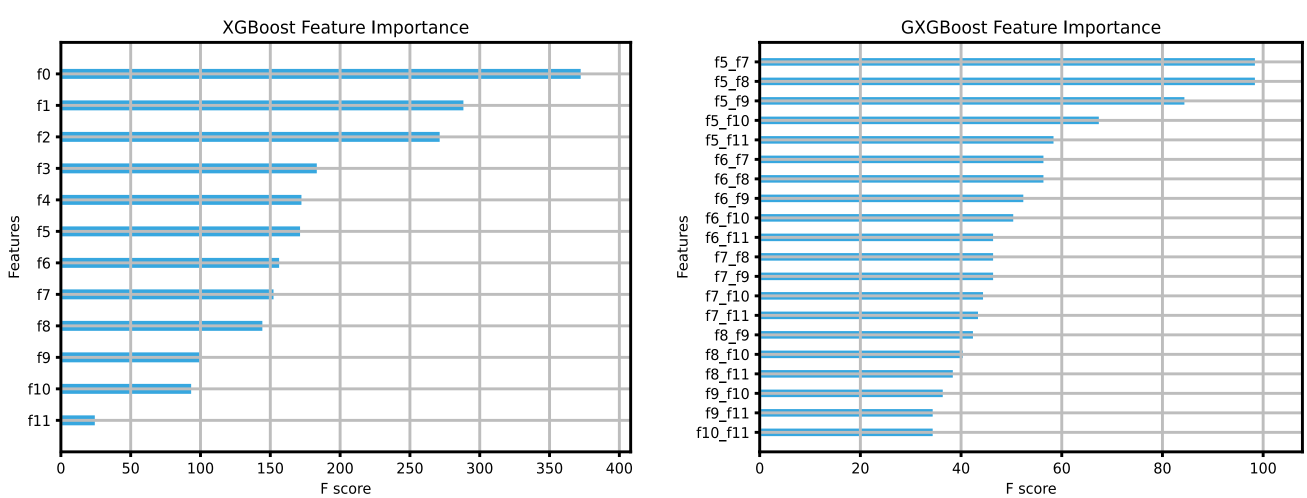

Fig. 5 Feature importance

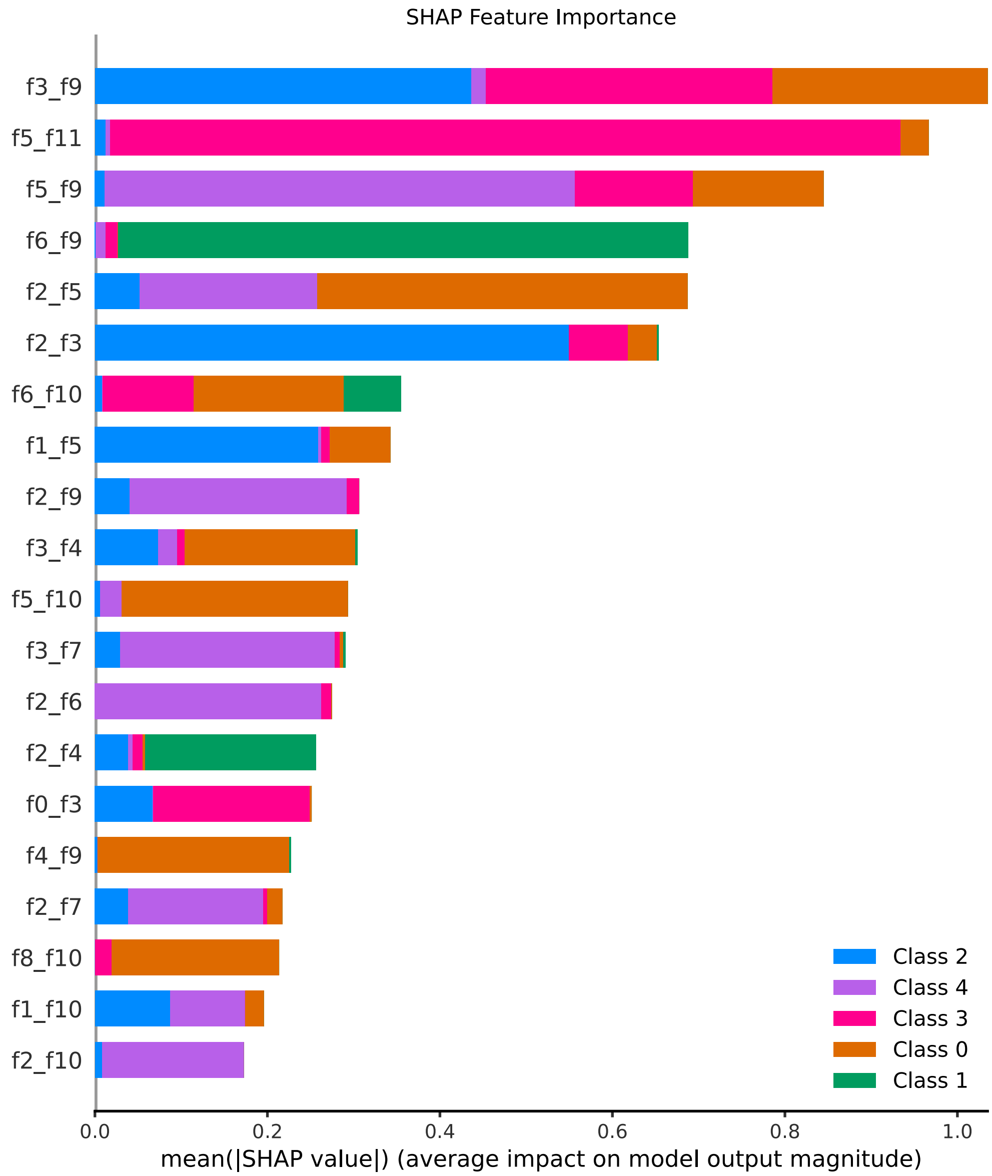

Fig. 6 Feature importance

Further, to enhance the interpretability of the model, we introduce the SHAP (SHapley Additive exPlanations) value analysis method to quantify the average impact of each feature on the prediction results from the perspective of model output. Figure 6 illustrates the results of the SHAP analysis. The most representative features include f3_f9, f5_f11, f5_f9, f6_f9, etc., and the distribution spans across multiple categories of outputs, verifying the significance of these granular features.In the statistical analysis, among the top 20 combinations of SHAP value importance, f2 appeared 7 times, followed by f3, f5, f9 and f10 (5 times each), and f4 and f6 (3 times each).Compared with the statistical results of GXGBoost, the SHAP analysis demonstrates a wider participation of raw features, especially the higher participation of the first segment features such as f2 and f3, revealing that the granular features not only encode the interaction information, but also maintain a certain degree of interpretability of the raw features.

In summary, although GXGBoost introduces the granularity mechanism to increase the model complexity, it captures the nonlinear interaction information between features more effectively by the way of feature combination.Combined with the results of SHAP analysis, we find that the

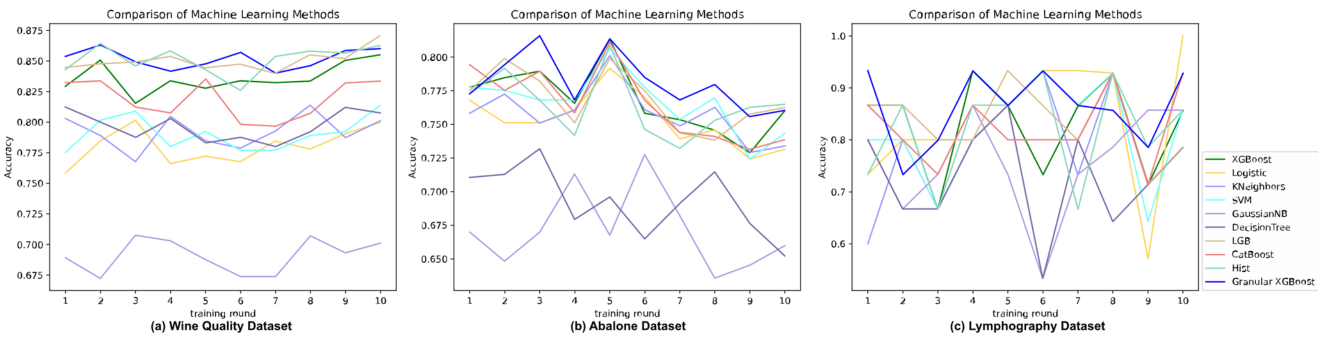

Fig. 7 Comparison of machine learning methods

key granularity features of GXGBoost and the high-impact features of SHAP have a high degree of overlap, which indicates that GXGBoost retains the advantage of XGBoost's interpretability, and at the same time provides a more fine-grained basis for discrimination through feature granularity. Therefore, GXGBoost achieves a better balance between model transparency and practicality.

5.5 Comparison of classification algorithms

This experiment compares the proposed granular XGBoost classification algorithm, GXGBoost, with other classification algorithms, including XGBoost, Logistic Regression, K-Nearest Neighbors (KNeighbors), Support Vector Machine (SVM), Gaussian Naive Bayes (GaussianNB), Decision Tree, LightGBM, CatBoost, and Histogram-Based Gradient Boosting (HistGB) across 23 datasets. Accuracy serves as the evaluation metric to ensure the reliability and comparability of the results. To enhance the accuracy and robustness of the findings, each dataset was tested 10 times. These multiple runs help mitigate the impact of random factors and yield more reliable evaluation results. Figure 7 illustrates the accuracy of each machine learning method over multiple runs, offering a detailed comparison of the performance of different algorithms on the Wine Quality, Abalone, and Lymphography datasets.

As illustrated in Fig. 7(a), the GXGBoost classification algorithm outperforms the other nine machine learning methods in most training rounds, achieving the best classification results overall. While the Histogram-Based Gradient Boosting (HistGB) algorithm excelled in certain iterations, GXGBoost consistently delivered superior results in the remaining iterations. In Fig. 7(b), GXGBoost demonstrates higher accuracy across the majority of iterations when compared to other classification algorithms. Furthermore, Fig. 7(c) indicates that GXGBoost exhibits less fluctuation in accuracy than the other classification algorithms.

Since some of the benchmark datasets used in this study (such as Haberman and Pima Indians) exhibit significant class imbalance, we evaluate the performance of GXGBoost not only using accuracy, but also F1-score. The F1-score is particularly useful in imbalanced classification tasks as it accounts for both precision and recall, providing a more informative measure of model performance on minority classes. To further validate the reliability and accuracy of the results, we evaluated both the classification accuracy and F1-Score of various machine learning methods on each dataset using a ten-fold cross-validation. This approach the calculation of the mean and variance of these metrics for each algorithm, ensuring the robustness and credibility of the evaluation process. The accuracy rates of the classification algorithms are presented in Table 5, while the F1-Scores are displayed in Table 6. The results are reported to six decimal places to facilitate a more precise comparison. Bold values in Tables 5 and 6 indicate the highest accuracy achieved by GXGBoost on each dataset.

When using accuracy as a performance evaluation metric, Table 5 demonstrates that the GXGBoost classification algorithm outperforms other commonly utilized classification algorithms across the majority of datasets. Although the accuracy of GXGBoost is slightly lower than that of certain classification algorithms on the Iris, Haberman's Survival, Heart Disease, Ionosphere, Wholesale Customers, and Blood Transfusion Service Center datasets, it shows a significant advantage in the remaining datasets. Specifically, in the Iris dataset, the GXGBoost algorithm performs marginally better than the XGBoost algorithm, yet it underperforms compared to the K-Nearest Neighbors classifier. In the Heart Disease dataset, the accuracy score of the GXGBoost algorithm is only slightly lower than that of the highest-scoring Logistic algorithm, while demonstrating a 2.5926%

Table 5 Comparison of Classification Algorithm Accuracy Across Datasets

Dataset	GXGBoost	XGBoost	Logistic	KNeighbors	SVM	GaussianNB	DecisionTree	LightGBM	CatBoost	HistGB
Iris	0.966667± 0.002869	0.933333± 0.003556	0.933333± 0.004844	**0.980000± 0.001822**	0.960000± 0.001956	0.953333± 0.001822	0.946667± 0.003378	0.946667± 0.000678	0.943333± 0.003600	0.940000± 0.003956
Wine	**0.983007± 0.000675**	0.949673± 0.002752	0.981889± 0.000494	0.955556± 0.001728	0.981689± 0.001111	0.977778± 0.001975	0.916013± 0.003206	0.982070± 0.000675	0.966340± 0.002608	0.972222± 0.002623
Wine Quality	**0.855781± 0.000208**	0.832078± 0.000182	0.7808516± 0.000457	0.792524± 0.000332	0.792368± 0.000456	0.692320± 0.000222	0.794524± 0.000175	0.849472± 0.000177	0.818997± 0.000343	0.841470± 0.000827
Haberman's Survival	0.734731± 0.008496	0.675806± 0.010019	0.735161± 0.007397	0.718495± 0.006473	0.722043± 0.011166	0.744946± 0.005379	0.630323± 0.008054	0.701613± 0.010756	**0.751183± 0.009371**	0.701720± 0.012669
Heart Disease	0.833333± 0.002538	0.807407± 0.002414	**0.844444± 0.008176**	0.803704± 0.004129	0.822222± 0.004609	0.8148148± 0.000837	0.755556± 0.004719	0.792593± 0.002524	0.8271605± 0.001701	0.807407± 0.002140
Breast Cancer Coimbra	**0.785606± 0.009883**	0.697727± 0.012023	0.637121± 0.016844	0.733333± 0.011717	0.739394± 0.009782	0.631061± 0.019456	0.748485± 0.019167	0.750758± 0.012838	0.707576± 0.016726	0.757576± 0.011788
Titanic	**0.823271± 0.000603**	0.809238± 0.001582	0.795793± 0.002111	0.799126± 0.000909	0.805893± 0.002245	0.795793± 0.002187	0.751985± 0.001136	0.804782± 0.001762	0.817104± 0.001042	0.807016± 0.001582
Pima Indians	**0.759108± 0.002613**	0.735646± 0.002749	0.755297± 0.002826	0.740772± 0.001836	0.743472± 0.002573	0.753930± 0.003687	0.716063± 0.005151	0.746053± 0.002294	0.753930± 0.001622	0.746087± 0.001260
Abalone	**0.786692± 0.000164**	0.771142± 0.000347	0.751506± 0.000384	0.753180± 0.000255	0.769699± 0.000436	0.670818± 0.000373	0.698838± 0.000337	0.768742± 0.000323	0.769459± 0.000216	0.764668± 0.000191
Balance Scale	**0.907220± 0.002094**	0.847977± 0.000679	0.867051± 0.001523	0.862366± 0.001092	0.905556± 0.000641	0.907143± 0.000817	0.790323± 0.000763	0.875243± 0.000490	0.883129± 0.000833	0.839964± 0.000934
Breast Tissue	**0.745455± 0.021537**	0.670000± 0.010827	0.527273± 0.014760	0.630000± 0.014827	0.536364± 0.010876	0.652727± 0.030625	0.677273± 0.011872	0.716364± 0.017749	0.687273± 0.022185	0.660909± 0.011488
Sonar	**0.850952± 0.005548**	0.821190± 0.005220	0.764286± 0.005052	0.802619± 0.003211	0.812143± 0.003345	0.661190± 0.011596	0.705238± 0.004474	0.845238± 0.004689	0.840714± 0.005147	0.840238± 0.008995
Energy Efficiency	**0.940123± 0.000613**	0.901059± 0.000914	0.779887± 0.003147	0.809826± 0.002296	0.790294± 0.003049	0.727768± 0.003074	0.937543± 0.000361	0.916695± 0.001352	0.925837± 0.000637	0.927102± 0.000511
HCV	**0.944791± 0.000691**	0.930196± 0.000945	0.900714± 0.001100	0.892649± 0.001282	0.900793± 0.000986	0.907456± 0.005367	0.907403± 0.001675	0.936674± 0.000649	0.935008± 0.000737	0.939926± 0.000781
IndianLiverPatientDataset	**0.720281± 0.002498**	0.689568± 0.001843	0.713501± 0.001375	0.708153± 0.004783	0.713355± 0.003067	0.554179± 0.001626	0.675716± 0.001263	0.689421± 0.002169	0.709994± 0.002514	0.697954± 0.003088
Ionosphere	0.934286± 0.001967	0.922857± 0.000824	0.880000± 0.001110	0.845714± 0.001829	0.937143± 0.002416	0.885714± 0.001796	0.877143± 0.001804	**0.937143± 0.001110**	0.937143± 0.001600	0.928571± 0.001347
Parkinsons	**0.944211± 0.002219**	0.929737± 0.006090	0.857895± 0.008008	0.832368± 0.008406	0.816316± 0.006057	0.701842± 0.004977	0.882368± 0.004115	0.938947± 0.002435	0.934737± 0.006018	0.934211± 0.003003
Seeds	**0.933333± 0.002812**	0.919048± 0.004104	0.923810± 0.001905	0.923810± 0.002812	0.919048± 0.001383	0.895238± 0.005351	0.900000± 0.002925	0.928571± 0.002834	0.928571± 0.002834	0.914286± 0.003084
TeachingAssistantEvaluation	**0.680000± 0.016711**	0.626667± 0.006400	0.546667± 0.006933	0.486667± 0.012489	0.520000± 0.006044	0.520000± 0.014044	0.646667± 0.010711	0.526667± 0.013733	0.626667± 0.011733	0.533333± 0.007111
Lymphography	**0.884286± 0.007728**	0.850952± 0.005175	0.829524± 0.004961	0.803333± 0.005814	0.796667± 0.007148	0.756190± 0.024571	0.734762± 0.001234	0.836190± 0.008391	0.835714± 0.008695	0.788571± 0.010554
Vertebral Column	**0.858065± 0.006493**	0.812903± 0.008907	0.848387± 0.005630	0.832258± 0.003913	0.848387± 0.006878	0.774194± 0.006868	0.803226± 0.004880	0.822581± 0.006920	0.845161± 0.005161	0.812903± 0.005994

Table 5 (continued)

Dataset	GXGBoost	XGBoost	Logistic	KNeighbors	SVM	GaussianNB	DecisionTree	LightGBM	CatBoost	HistGB
Wholesale customers	0.718182± 0.001880	0.688636± 0.003001	0.718182± 0.003946	0.647727± 0.001989	**0.718182± 0.001157**	0.459091± 0.013202	0.545455± 0.004442	0.672727± 0.001880	0.693182± 0.000852	0.656818± 0.002216
BloodTransfusionService-Center	0.772306± 0.000736	0.761568± 0.001674	0.771027± 0.002084	0.769568± 0.001448	0.771027± 0.001650	0.752234± 0.001600	0.708126± 0.000695	0.740198± 0.001021	**0.778991± 0.001359**	0.741495± 0.000863

improvement in accuracy over the XGBoost classification algorithm. In the remaining four datasets, the GXGBoost algorithm is slightly lower than some of the machine learning algorithms, yet still surpasses the XGBoost algorithm. In 17 datasets, including Wine and Wine Quality, the accuracy of GXGBoost is significantly superior to that of the other nine classification algorithms. Furthermore, in the Breast Cancer Coimbra dataset, the GXGBoost algorithm enhances accuracy by 8.7879% compared to the XGBoost algorithm. Similarly, in the Breast Tissue dataset, the GXGBoost algorithm achieves an accuracy improvement of 7.5455% over the XGBoost algorithm.

From Table 6, when utilizing the F1-Score as a performance evaluation metric, the F1-Scores of the GXGBoost algorithm are significantly higher than those of the other nine algorithms across 19 out of 23 datasets, excluding the Wine Quality, Heart Disease, Ionosphere, and Vertebral Column datasets. In the cases of the Wine Quality, Heart Disease, Ionosphere, and Vertebral Column datasets, the F1-Scores of the GXGBoost algorithm also surpass those of the other eight algorithms, including the XGBoost algorithm. On the Haberman dataset, which is known for its class imbalance, GXGBoost achieves a significantly higher F1-score compared to standard XGBoost, indicating better minority class prediction. This improvement suggests that the multi-distance granular representation helps the model capture patterns in the underrepresented class more effectively. Similar trends are observed on the Pima Indians dataset, further validating the robustness of GXGBoost against imbalanced distributions.

6 Conclusion

The proposed granular XGBoost classifier, GXGBoost, enhances dimensionality through pairwise feature combinations and employs multiple distance metric methods to granulate data, thereby generating multi-level feature representations. These representations are subsequently input into the GXGBoost classifier for training, which effectively improves classification precision. This method fully leverages the efficiency, robustness, and high accuracy of XGBoost, combined with the innovative design of multi-distance metric granulation methods, significantly enhancing the model's feature representation capability and predictive performance. Experimental results validate the feasibility of this approach and demonstrate its potential in addressing overfitting issues in small sample datasets characterized by uneven categories, sparse data, and noise influences. Future work will focus on further optimizing the granulation method and exploring its applications and extensions in other machine learning tasks.

Table 6 Comparison of classification algorithm F1-Score across datasets

Dataset	GXGBoost	XGBoost	Logistic	KNeighbors	SVM	GaussianNB	DecisionTree	LightGBM	CatBoost	HistGB
Iris	**0.966934± 0.002822**	0.949680± 0.04376	0.926710± 0.004063	0.965208± 0.002345	0.959810± 0.004739	0.959691± 0.003906	0.947048± 0.003318	0.946649± 0.004374	0.953887± 0.002640	0.946476± 0.005267
Wine	**0.993964± 0.000328**	0.960535± 0.001927	0.989072± 0.000478	0.953963± 0.001966	0.988527± 0.000530	0.977225± 0.000781	0.915180± 0.002040	0.972310± 0.000767	0.983449± 0.000640	0.972299± 0.000768
Wine Quality	0.829517± 0.000536	0.811971± 0.000539	0.733191± 0.000564	0.783407± 0.000344	0.739533± 0.000667	0.706973± 0.000241	0.797269± 0.000484	**0.832233± 0.000215**	0.803963± 0.000650	0.832219± 0.000214
Haberman's Survival	**0.709272± 0.003009**	0.672512± 0.005725	0.630147± 0.013748	0.654021± 0.010961	0.616827± 0.011159	0.678881± 0.014465	0.684820± 0.003102	0.688997± 0.004422	0.704000± 0.007233	0.686180± 0.006126
Heart Disease	0.837479± 0.005330	0.815336± 0.003838	0.833294± 0.004084	0.814334± 0.003085	0.832603± 0.003474	**0.851876± 0.003190**	0.773728± 0.004995	0.803569± 0.002485	0.843791± 0.003398	0.800032± 0.004002
BreastCancer-Coimbra	**0.739258± 0.011543**	0.652553± 0.011543	0.587241± 0.007730	0.689344± 0.015369	0.728326± 0.018315	0.581311± 0.019019	0.675460± 0.015331	0.715595± 0.005074	0.720382± 0.012940	0.686224± 0.006021
Titanic	**0.816741± 0.002028**	0.802373± 0.002719	0.794228± 0.001658	0.785334± 0.002127	0.801137± 0.002156	0.794368± 0.001492	0.751904± 0.003280	0.808608± 0.002466	0.809753± 0.003079	0.815510± 0.001914
Pima Indians	**0.766345± 0.001794**	0.748849± 0.001603	0.749859± 0.002360	0.706406± 0.001736	0.759612± 0.003130	0.746210± 0.004374	0.695519± 0.001727	0.762266± 0.001192	0.758688± 0.002237	0.741747± 0.000780
Abalone	**0.771681± 0.000465**	0.758863± 0.000390	0.739485± 0.000051	0.741202± 0.000526	0.755953± 0.000171	0.678186± 0.000077	0.688904± 0.000114	0.761222± 0.000407	0.756241± 0.000635	0.760318± 0.000254
Balance Scale	**0.899604± 0.004141**	0.840276± 0.002418	0.828919± 0.003543	0.845040± 0.002155	0.868457± 0.002757	0.869814± 0.002853	0.807160± 0.001547	0.861194± 0.001870	0.894999± 0.001975	0.845812± 0.002108
Breast Tissue	**0.720203± 0.013499**	0.709229± 0.013495	0.483455± 0.021760	0.651485± 0.011014	0.511646± 0.014547	0.651844± 0.017736	0.647998± 0.028728	0.664357± 0.020826	0.671429± 0.012672	0.683147± 0.025577
Sonar	**0.882787± 0.007770**	0.868393± 0.006828	0.779273± 0.010071	0.841400± 0.008169	0.833099± 0.004328	0.683241± 0.008097	0.731755± 0.007374	0.869862± 0.005582	0.857935± 0.006003	0.860500± 0.005314
Energy Efficiency	**0.939140± 0.001630**	0.900366± 0.001375	0.751158± 0.005233	0.777705± 0.003498	0.756806± 0.004642	0.652610± 0.003928	0.918713± 0.001685	0.928139± 0.001193	0.935863± 0.001266	0.937788± 0.001672
HCV	**0.933876± 0.000770**	0.920713± 0.000769	0.860548± 0.002687	0.849994± 0.003219	0.864898± 0.002677	0.910988± 0.002420	0.913182± 0.000795	0.932356± 0.000976	0.932760± 0.000862	0.931648± 0.001245
IndianLiverPatientDataset	**0.700027± 0.001807**	0.661007± 0.005437	0.651729± 0.004022	0.653157± 0.003695	0.595464± 0.005599	0.559262± 0.008174	0.633014± 0.001839	0.676683± 0.002978	0.651496± 0.007481	0.669233± 0.004725
Ionosphere	0.935665± 0.002355	0.921238± 0.002070	0.866608± 0.003204	0.850944± 0.002869	**0.938440± 0.001974**	0.887247± 0.002658	0.890512± 0.000260	0.937829± 0.002258	0.932129± 0.002989	0.928958± 0.002205
Parkinsons	**0.937697± 0.003162**	0.906658± 0.004611	0.856645± 0.004515	0.831182± 0.005577	0.775379± 0.013973	0.734236± 0.007952	0.846953± 0.008012	0.919299± 0.004361	0.934536± 0.004132	0.918635± 0.006517
Seeds	**0.933420± 0.0041788**	0.908076± 0.003096	0.918773± 0.001811	0.9188135± 0.003207	0.927555± 0.001492	0.898637± 0.003027	0.900704± 0.001964	0.927673± 0.002081	0.925897± 0.002622	0.903467± 0.003324
TeachingAssistantEvaluation	**0.676574± 0.021236**	0.630717± 0.015166	0.523702± 0.007578	0.490618± 0.020868	0.526239± 0.004558	0.477410± 0.012809	0.667031± 0.018883	0.500375± 0.018163	0.615878± 0.023185	0.530137± 0.023596
Lymphography	**0.852283± 0.004185**	0.835501± 0.011557	0.827306± 0.005736	0.765624± 0.018167	0.810067± 0.011656	0.704479± 0.031860	0.769614± 0.007128	0.848432± 0.003617	0.848107± 0.004561	0.820095± 0.011209
Vertebral Column	0.843214± 0.003669	0.813420± 0.004396	0.845489± 0.004229	0.823273± 0.002358	0.840652± 0.005131	0.781983± 0.003719	0.804213± 0.002736	0.805126± 0.003294	**0.846262± 0.003295**	0.817309± 0.002533

Table 6 (continued)

Dataset	GXGBoost	XGBoost	Logistic	KNeighbors	SVM	GaussianNB	DecisionTree	LightGBM	CatBoost	HistGB
Wholesale customers	**0.613761±0.005062**	0.596142±0.005521	0.601837±0.006277	0.575201±0.005094	0.601837±0.006277	0.476805±0.008085	0.556190±0.002801	0.605315±0.007324	0.600096±0.005367	0.597378±0.007138
BloodTransfusionServiceCenter	**0.764153±0.004129**	0.747589±0.005103	0.690472±0.004143	0.731334±0.004350	0.694039±0.005511	0.715238±0.006193	0.717476±0.008008	0.738656±0.004090	0.762605±0.003508	0.733792±0.005549

6.1 Abbreviations

- CatBoost: CategoricalBoosting
- CNN: Convolutional Neural Network
- DCNN: Deep Convolutional Neural Network
- GXGBoost: Granular XGBoost
- HistGB: Hist Gradient Bossting
- KNeighbors: K-Nearest Neighbors
- LightGBM: Light Gradient Boosting Machine
- Logistic: logistic regression
- SMOTE: Synthetic Minority Over-sampling Technique
- SVM: Support Vector Machine
- XGBoost: eXtreme Gradient Boosting

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Data Availability and Access The data used to support the findings of this study is available from the corresponding author upon request.

Declarations

Ethical and Informed Consent for Data Used The data covered in this article are from the UCI Open Data Set, so informed consent is not required for the inclusion of data in the manuscript.

Conflicts of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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